Planar Steiner diversities need logarithmic ℓ_1 distortion

Jeff Kline

Pre-release draft — 2026-10-25

Pre-release draft. Not admitted, not peer reviewed.

Abstract

The Steiner diversity of a graph assigns to each vertex set the minimum cost of a connected subgraph containing it. On unit diamond graphs, this diversity requires logarithmic distortion in ℓ_1, although the graphs are planar and have treewidth two. After $d \geq 1$ edge-replacement rounds the graph has $N_d = (2 \cdot 4^d + 4)/3$ vertices, and we prove the explicit lower bound $c_1(D_d) \geq (6d + 7 + 2/4^d)/9$. A cut inequality on the nested generation sets, together with their exact Steiner costs, gives a rational dual certificate with only d demand sets. Combined with Bryant and Tupper's upper bound for metric Steiner diversities, the result has order $\log N_d$. The same certificate supplies unit-capacity set demands that satisfy every cut condition but require at least this much routing congestion. The exact value of $c_1(D_d)$ is not determined for $d \geq 3$. Three bounded literature scans found no matching result; novelty remains unresolved.

1 Introduction

The Steiner diversity of a graph assigns to each vertex set the minimum cost of a connected subgraph containing it. On unit diamond graphs, this diversity requires logarithmic distortion in ℓ_1, although the graphs are planar and have treewidth two. The obstruction concerns one embedding that must preserve the connection costs of all terminal sets simultaneously.

A diversity extends a metric from pairs to finite sets. For a finite set V, it is a function $\delta : 2^V \to [0, \infty)$ that vanishes exactly on sets of size at most one and satisfies

$$\delta(A \cup B) \leq \delta(A \cup C) + \delta(C \cup B) \qquad (C \neq \varnothing).$$

Bryant and Tupper introduced this framework [1] and developed its connection to hypergraph cuts and flows [2]. For a finite connected graph $G = (V, E)$ with positive edge lengths ℓ, its *graph Steiner diversity* is

$$\delta_{G,\ell}(A) = \min \left\{ \sum_{e \in E(H)} \ell(e) : H \subseteq G \text{ is connected and } A \subseteq V(H) \right\},$$

with value zero for $|A| \leq 1$. Every vertex of G is available as a Steiner point. Edges are ordinary pairs throughout this paper.

For $f : V \to \mathbb{R}^m$, the coordinate-range ℓ_1 diversity is

$$h_f(A) = \sum_{i=1}^{m} \left(\max_{v \in A} f_i(v) - \min_{v \in A} f_i(v) \right),$$

again zero on the empty set. Write $c_1(\delta)$ for the infimum of $D \geq 1$ over all finite dimensions and maps satisfying $\delta(A) \leq h_f(A) \leq D\delta(A)$ for every $A \subseteq V$. Rescaling the map permits this noncontracting normalization. For unit graph lengths we write $c_1(G)$.

Recent metric results in OpenAI's Erdős problem #089 materials concern constant-distortion embeddings for planar graphs and graphs of fixed treewidth [5, 6]. These are statements about pairs. For the series-parallel class containing diamonds, the bound of two is attributed to Chakrabarti, Jaffe, Lee and Vincent [7], as reported in [5, Section 1.2]. We use that attribution only as context; we do not claim that diamond metrics embed isometrically. The question here is whether a constant can also preserve every Steiner set cost. The answer is no, already at treewidth two.

Results and scope. The principal result is the explicit all-depth bound in Theorem 1. Its proof uses a simultaneous cut inequality on d nested sets, rather than pairwise separation. Corollaries 4 and 5 give the asymptotic order and a normalized set-demand flow–cut consequence. Secondary results identify the exact optimum of the restricted certificate method and its dependence on edge lengths. None of these statements determines the full optimal distortion at depth three or beyond. The computational table checks finite controls; it is not a premise of the proof. Prior work is credited in Section 5. The three recorded searches were bounded and source-negative, so novelty remains unresolved.

2 The diamond lower bound

Let D_0 be one edge. To obtain D_d from D_{d-1}, replace every edge by two internally disjoint paths of two edges each. Every final edge has unit length. Retain all earlier vertices, and let V_j be the vertex set present after generation j, viewed as a subset of V_d.

Theorem 1. *For every integer $d \geq 1$, the graph D_d has 4^d edges and $N_d = (2 \cdot 4^d + 4)/3$ vertices. It is planar and has treewidth exactly two. Its graph Steiner diversity satisfies*

$$c_1(D_d) \geq L_d := \frac{6d + 7 + 2/4^d}{9}.$$

Consequently, neither planar graphs nor graphs of treewidth two admit a uniform constant bound on ℓ_1 diversity distortion.

2.1 Cut representation and duality

For a cut $S \subseteq V$, let $\chi_S(A)$ be one if A meets both S and $V \setminus S$, and zero otherwise. Every coordinate-range ℓ_1 diversity on a finite set is a finite nonnegative sum

$$h(A) = \sum_S w_S \chi_S(A), \qquad w_S \geq 0.$$

Indeed, sort the distinct values of each coordinate and assign each consecutive gap to its upper threshold cut. A set crosses exactly the thresholds between its minimum and maximum. Conversely, the coordinate $w_S \mathbf{1}_S$ realizes $w_S \chi_S$.

Suppose nonnegative measures α, β on subsets of V satisfy

$$\sum_A \alpha_A \chi_S(A) \leq \sum_A \beta_A \chi_S(A) \quad \text{for every } S, \qquad \sum_A \beta_A \delta(A) = 1. \tag{1}$$

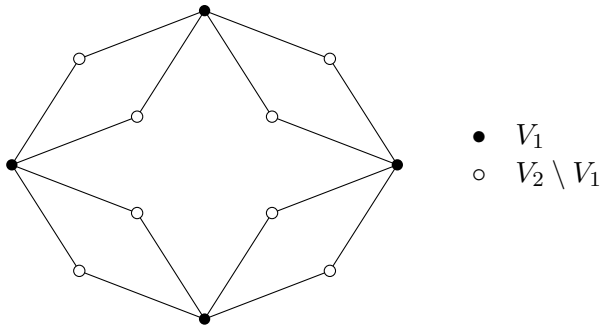


Figure 1: The unit diamond D_2. All twelve vertices form V_2; the four filled vertices form V_1. Their Steiner costs are respectively 11 and 6. The drawing does not encode edge lengths.

Then any $\delta \le h \le D\delta$ obeys

$$\sum_A \alpha_A \delta(A) \le \sum_A \alpha_A h(A) \le \sum_A \beta_A h(A) \le D.$$

This is the dual-certificate principle used below. It applies in every allowed finite dimension.

2.2 Geometry and graph class

Each replacement multiplies the number of edges by four and adds two vertices per old edge. Thus

$$E_d = 4^d, \qquad N_d = 2 + 2\sum_{i=0}^{d-1} 4^i = \frac{2 \cdot 4^d + 4}{3}.$$

Each edge of D_j has a descendant gadget isomorphic to D_{d-j}. Distinct gadgets have disjoint edge sets and disjoint interiors, meeting only at old endpoints. Within a gadget the terminal distance is 2^{d-j}, and there are 2^{d-j} edge-disjoint terminal paths. Both assertions follow inductively: two smaller diamonds occur in series in each of two parallel branches. Distances double; corresponding paths concatenate in each branch, and the two branch collections are edge-disjoint.

Every edge of D_j, for $j \ge 1$, lies in the four-cycle replacing its parent edge. Hence D_j is connected and bridgeless, and each nontrivial cut crosses at least two edges.

Replacing each edge in a planar drawing inside a narrow neighborhood preserves planarity. For a width-two tree decomposition, start with the bag containing the original two endpoints. When xy is replaced through new vertices a, b, attach bags $\{x, y, a\}$ and $\{x, y, b\}$ to an existing bag containing x, y. Old edge coverage ensures such a bag exists. The new edges are covered, and the bags containing each vertex remain connected. Retain old bags after their edges disappear. Every bag has size at most three. Since D_d contains a cycle for $d \ge 1$, its treewidth is exactly two.

2.3 A simultaneous inequality for every cut

Set

$$a_j = 2^{d-j} \quad (1 \le j < d), \qquad a_d = 2.$$

Lemma 2. *For every cut S of D_d,*

$$\sum_{j=1}^{d} a_j \chi_S(V_j) \le |\partial_{D_d} S|. \tag{2}$$

Proof. Trivial cuts have both sides zero. For a nontrivial cut, let q be the first generation in $\{1, \dots, d\}$ with $\chi_S(V_q) = 1$. It exists since V_d is the whole vertex set. By nesting, the indicators vanish before q and are one thereafter. The restriction to V_q is a nontrivial cut of D_q, so it crosses at least two coarse edges. Their descendant gadgets are edge-disjoint. In each, S separates the endpoints and therefore crosses all 2^{d-q} edge-disjoint terminal paths. Consequently

$$|\partial_{D_d} S| \geq 2^{d-q+1} = \sum_{j=q}^{d} a_j = \sum_{j=1}^{d} a_j \chi_S(V_j).$$

This includes $q = d$ and $d = 1$. □

2.4 Exact Steiner costs of the generation sets

Lemma 3. *For $1 \leq j \leq d$,*

$$\delta_{D_d}(V_j) = 2^{d-j}(N_j - 1). \tag{3}$$

Proof. For the upper bound, choose a spanning tree of D_j and a shortest terminal path in each of its $N_j - 1$ edge gadgets. Their union is connected, contains V_j, and has $2^{d-j}(N_j - 1)$ edges.

For the lower bound, let H be any connected subgraph containing V_j. Retain a coarse edge of D_j whenever H connects its endpoints within its descendant gadget. The retained edges connect V_j: a simple H-path between old vertices, divided at successive old vertices, consists of passages through single gadgets between their endpoints. Different gadgets meet only at old endpoints. Each passage gives a retained coarse edge. At least $N_j - 1$ gadgets are therefore retained. Each costs at least 2^{d-j} edges, and their edge sets are disjoint. This proves the lower bound even when H uses arbitrary newer Steiner vertices. □

2.5 The certificate and its objective

Put $\alpha_{V_j} = a_j/4^d$ and put $\beta_{\{u,v\}} = 1/4^d$ on every final graph edge, with all other coefficients zero. Every edge pair has Steiner cost one, so $\sum_A \beta_A \delta(A) = 1$. Dividing (2) by 4^d gives every cut constraint in (1). By (3),

$$\begin{aligned}
\sum_A \alpha_A \delta(A) &= \frac{1}{4^d}\left[\sum_{j=1}^{d-1} \frac{4^{d-j}(2 \cdot 4^j + 1)}{3} + \frac{2(2 \cdot 4^d + 1)}{3}\right] \\
&= \frac{2(d-1)}{3} + \frac{1}{9}\left(1 - 4^{-(d-1)}\right) + \frac{4}{3} + \frac{2}{3 \cdot 4^d} \\
&= \frac{6d + 7 + 2/4^d}{9}.
\end{aligned}$$

The sum is empty when $d = 1$. The dual-certificate principle proves Theorem 1. Since N_d is proportional to 4^d, $L_d = (2/3)\log_4 N_d + O(1)$, which diverges.

3 Consequences and limits

Corollary 4. *The unit diamond diversities have $c_1(D_d) = \Theta(\log N_d)$. The lower bound is self-contained; the upper bound uses [3, Theorem 2].*

Proof. Bryant and Tupper prove an $O(\log n)$ upper bound for the Steiner diversity of an n-point metric space. Graph Steiner costs equal the Steiner costs in the shortest-path metric completion on the same vertex set. Replacing metric-tree edges by graph shortest paths gives one inequality; viewing a graph Steiner tree as a tree of metric edges gives the other. All vertices remain available as Steiner points. The cited theorem and Theorem 1 now give the claimed order. □

Corollary 5 (Set-demand flow–cut form)**.** *Give every edge of D_d capacity one and give terminal set V_j demand a_j, with all other demands zero. These demands satisfy every cut condition, but any fractional Steiner-tree routing of the full demands requires congestion at least L_d. In the notation of [2, Section 5],*

$$\mathrm{MinHypCut} = 1, \qquad \mathrm{MaxHSP} \le L_d^{-1}, \qquad \frac{\mathrm{MinHypCut}}{\mathrm{MaxHSP}} \ge L_d.$$

Proof. The cut condition is exactly (2). The cut ratio is boundary capacity divided by $\sum_j a_j \chi_S(V_j)$, for nonzero denominator. Every such ratio is at least one. A singleton vertex born in generation d has degree two and crosses only the demand set V_d, whose demand is two; its ratio is one.

For each j, distribute mass a_j over trees containing V_j. The sum of all edge loads is at least $\sum_j a_j \delta(V_j) = 4^d L_d$. Congestion ρ with unit capacities bounds that sum above by $\rho 4^d$, hence $\rho \ge L_d$. Bryant and Tupper's maximum concurrent packing variable f routes mass $f a_j$ under the original capacities. The same calculation gives $f 4^d L_d \le 4^d$. This verifies the stated normalization directly. Their Theorems 1–2 and 17–18 establish the general diversity/flow–cut correspondence; the calculation here supplies these particular demands. □

Proposition 6 (Secondary: length dependence)**.** *Common positive rescaling of all edge lengths preserves c_1. If the lengths on D_d satisfy $0 < m \le \ell(e) \le M$ and $R = M/m$, then*

$$c_1(D_d, \ell) \ge \max\{1, L_d/R\}.$$

There are also strictly positive length assignments on every underlying diamond for which $c_1 = 1$.

Proof. Common rescaling multiplies all Steiner costs and all embedding coordinates by the same factor. For bounded lengths, minimizing over the same connected subgraphs gives $m\delta_{\mathrm{unit}} \le \delta_\ell \le M\delta_{\mathrm{unit}}$. If $\delta_\ell \le h \le D\delta_\ell$, then h/m is noncontracting for δ_{unit} with distortion at most DR. Take infima and apply Theorem 1.

For the last assertion, choose a spanning tree T on N vertices, give its edges length one, and give every other edge length N. Every non-tree edge can be replaced by its T-path of at most $N-1$ edges. Thus every connected subgraph spanning A can be replaced by one in T of no greater cost, and $\delta_{G,\ell}(A) = \delta_T(A)$ for every A. The tree diversity is the sum of its edge-cut diversities, so it embeds exactly in ℓ_1. □

Proposition 7 (Secondary: optimum of the restricted method)**.** *On the unit diamond, fix $\beta_{\{u,v\}} = 1/4^d$ on every edge and require α to be supported on $V_1, \dots, V_d$. Among feasible nonnegative α in (1), the maximum objective is L_d, uniquely attained by the coefficients of Theorem 1.*

Proof. Write $x_j = 4^d \alpha_{V_j}$, $y_q = \sum_{j=q}^d x_j$, and $b_q = 2^{d-q+1}$. The all-cut feasible region is exactly

$$x_j \ge 0, \qquad y_q \le b_q \quad (1 \le q \le d).$$

Necessity follows from a singleton vertex born in generation q: its degree is b_q, and it crosses exactly the generation sets indexed at least q. For sufficiency, the proof of Lemma 2 bounds any cut first splitting V_q below by b_q boundary edges.

Graph	Exact c_1	Graph	Exact c_1
Path, 8 vertices	1	D_2	17/8
Star, 8 vertices	1	Fan, 5 vertices	8/5
K_3	4/3	Fan, 8 vertices	7/4
K_4	3/2	Wheel, 5 vertices	5/3
K_5	8/5	Wheel, 8 vertices	38/21
K_6	5/3	Grid 2×3	5/3
K_8	7/4	Grid 2×4	7/4
C_4	3/2	Grid 2×5	9/5
C_5	8/5	Grid 2×6	11/6
C_6	5/3	Grid 3×3	11/6
C_7	12/7	Grid 3×4	29/15
C_8	7/4	Weighted 2-tree, seed 17	4/3
C_{10}	9/5	Weighted 2-tree, seed 31	14/11
C_{12}	11/6	Weighted 2-tree, seed 53	26/19

Table 1: Exact certificate values. Fans and wheels count all vertices, including the hub. The path has edge lengths $1, 2, 3$ repeated, the star has lengths $1, \ldots, 7$, and each 2-tree has eight vertices and recorded integer lengths in $1, \ldots, 9$. All other edges have unit length.

Let $c_j = \delta(V_j)$. Lemma 3 gives $c_1 > 0$ and $c_j - c_{j-1} = 2^{d-j}(4^j - 1)/3 > 0$ for $j \geq 2$. Summation by parts yields

$$4^{-d} \sum_{j=1}^{d} c_j x_j = 4^{-d} \left(c_1 y_1 + \sum_{j=2}^{d} (c_j - c_{j-1}) y_j \right).$$

Every coefficient is positive, so the maximum requires $y_q = b_q$ for all q. These equalities are simultaneously feasible since b_q decreases. They give $x_j = b_j - b_{j+1} = 2^{d-j}$ for $j < d$ and $x_d = 2$, as claimed. □

This restricted optimality does not establish the full dual optimum or $c_1(D_d) = L_d$ for $d \geq 3$. Uniform lengths provide the witness family; Theorem 1 is not a lower bound for every positive-length diamond. The result concerns graph Steiner diversities with arbitrary graph vertices as Steiner points, and finite-dimensional coordinate-range ℓ_1 targets. It is not a claim about terminal-induced subgraphs, terminal-only Steiner trees, or the distortion of pairs alone.

4 Exact finite controls

Table 1 records 28 saved rational primal/dual certificates. The draft verifier constructs D_1, D_2, checks every cut, enumerates exact Steiner costs, checks the objective identity for $d = 1, \ldots, 12$, and checks all 28 certificates. No new linear programs are solved. On each saved graph, the verifier computes the minimum spanning-tree cost on every induced vertex set, then minimizes over supersets of each terminal set. This gives exact Steiner costs because any connected subgraph contains a spanning tree. It checks both bounds on every terminal set, all dual cut constraints, normalization, and equality of primal and dual objectives using rational arithmetic. These finite checks do not verify the all-depth theorem.

All 28 final saved solver records report optimal status, and the exact certificate checks remove reliance on floating-point tolerances for these values. Three initial full formulations had timed out;

the saved successful runs used an equivalent reduction to upper constraints on edges and retained all lower constraints. The present verifier checks every upper constraint as well, so it does not rely on that reduction. The equalities $c_1(D_1) = 3/2$ and $c_1(D_2) = 17/8$ match L_1, L_2; they do not justify extrapolating equality to larger depths.

5 Earlier work and unresolved priority

Bryant and Tupper supply the diversity framework [1], the cut representation and hypergraph flow–cut connection [2], and the $O(\log n)$ Steiner-diversity upper bound [3, Theorem 2]. We cite the last result as an external dependency; its random-tree argument has not been independently audited here. The arXiv record identifies a 2017 version, although the saved PDF prints an October 2021 date.

Jozefiak and Shepherd [4] discuss a general-graph logarithmic lower bound inherited from expander metrics. That argument does not establish the planar, treewidth-two statement here. Gupta, Newman, Rabinovich and Sinclair [8] prove logarithmic lower bounds for embeddings of series-parallel metrics into distributions of dominating trees. This is a nearby metric result with a different target. Their publisher abstract was checked; the full article was not inspected for this draft. The series-parallel ℓ_1 metric bound attributed to [7] was read in the OpenAI papers' discussion, not in the original conference paper.

The author's earlier paper [9] concerns hypergraph Steiner diversities and an affine covering-versus-hitting construction. Its hyperedges may contain more than two vertices. The present witness consists entirely of graph edges and uses nested diamond generations.

Three saved bounded scans searched diversity terminology, flow–cut terminology, and selected citing papers. They found no source containing the same planar graph-Steiner lower bound or its generation-set certificate. That negative outcome is not a priority certificate. Search coverage, access failures, citations inspected only at abstract level, and unresolved candidates are recorded in the repository's prior-work audit. Novelty remains unresolved, including at the level of the mechanism.

AI assistance and verification status

AI systems assisted with the research, adversarial examination, finite certificate checks, source extraction, and preparation of this draft. A fresh full proof audit and the owner's adjudication are preserved as unchanged evidence. An initial examiner's alternative argument was adopted before that audit. The records distinguish an unsuccessful attempt to break a claim from a subsequent proof review. Neither constitutes peer review or formal proof-assistant certification. The author is responsible for the mathematical claims and their presentation. This document is a pre-release draft, not an admission or publication decision.

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